

Product Requirements Document (PRD)

OpenScholar

Version: 1.0

Status: MVP Planning (Updated)

1. Product Overview

1.1 Vision

To build a global open-access academic research platform where students can publish, discover, and access research without institutional or financial barriers.

1.2 Mission

- Democratize access to academic knowledge
- Enable student-first research publishing
- Build a scalable academic knowledge network
- Replace outdated research platforms with modern systems

1.3 Problem Statement

- Research is restricted by paywalls
- Platforms require institutional email access
- Student research lacks visibility
- Existing platforms have poor usability
- No centralized academic repository in Bangladesh

1.4 Solution Summary

OpenScholar provides:

- Open-access research publishing
- Free reading and downloading
- Simple publishing workflow
- Modern UI/UX
- Scalable architecture

2. Goals and Success Metrics

2.1 Business Goals

- Launch MVP within 4–5 months
- Onboard initial university
- Expand to multiple universities within 1 year

2.2 Product Goals

- Paper publishing within 5 minutes
- Search response time under 2 seconds
- High readability experience
- Free access to all content

2.3 Success Metrics

- Monthly Active Users: 10,000+
 - Papers Published: 5,000+
 - Average Session Time: 5+ minutes
 - Upload Success Rate: 99%
 - Search Response Time: <2 seconds
 - Retention Rate: 40%+
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3. Target Users

Primary:

- Undergraduate students
- Graduate students
- Research scholars

Secondary:

- Faculty members
 - Universities
 - General researchers
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4. Feature Prioritization (MoSCoW)

4.1 Must Have (MVP Core)

Authentication

- Email/password login
- Email verification
- Password reset
- Role-based access control

User Profile

- Basic profile
- Publication list

Research Publishing

- PDF upload ($\leq 50\text{MB}$)
- Metadata input (title, abstract, keywords)
- University and department selection
- Draft saving
- Version control (mandatory for edits)
- Multi-author support
- Author ordering (1st, 2nd, etc.)
- Submission status management

Admin Moderation

- Approve/reject submissions
- Publish approved papers
- Moderation logging

Paper Viewing

- Paper details page
- PDF viewer
- Download functionality

Search and Discovery

- Keyword search
- Filters (university, department, year)

User Engagement

- Reactions (1 per user per paper)
- Comments
- Reply to comments (1-level nesting)
- Share (link/social)

Basic Analytics

- View tracking
- Download tracking

Core Non-Functional Requirements

- Page load under 3 seconds
 - Search response under 2 seconds
 - Secure authentication
 - HTTPS communication
 - 99% uptime
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4.2 Should Have

- Dashboard (user analytics, most viewed papers)
 - Citation export
 - Multilingual support (English & Bangla)
 - Enhanced search filters (category)
 - Security enhancements
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4.3 Could Have

- Focus reading mode
 - Bookmark/save papers
 - Notifications
 - Advanced analytics (author-level, ranking)
 - Caching and CDN
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4.4 Won't Have (MVP Excluded)

- DOI integration
- Peer review system
- AI recommendation engine

- Automated plagiarism detection
 - Paid subscription system
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5. Differentiation

- Clean, user-focused interface
 - Simplified publishing workflow
 - Open-access model
 - Student-centered design
 - Engagement-driven academic platform
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6. User Flow

Publishing Flow:

Register → Login → Submit Paper → Admin Review → Publish

Discovery Flow:

Search → Filter → Open Paper → Read → Download → Interact (react/comment/share)

7. System Architecture

- Frontend: Next.js
 - Backend: API-based architecture
 - Database: PostgreSQL
 - Storage: Cloud storage (S3 or equivalent)
 - Authentication: JWT/session-based
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8. Non-Functional Requirements

Must:

- Performance: <3 seconds load time
- Security: encryption and HTTPS
- Reliability: 99% uptime

Should:

- Scalability: multi-university support
- Maintainability: modular architecture

Could:

- Zero-downtime deployment
 - Advanced monitoring
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9. Constraints

- Limited moderation resources
 - Initial deployment limited to one university
 - No DOI integration in MVP
 - No automated plagiarism detection
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10. Risks and Mitigation

- Fake submissions → Admin moderation
 - Plagiarism → Manual review
 - Low adoption → University onboarding
 - System overload → Cloud scaling
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11. Roadmap

Phase 1 (MVP):

- Authentication
- Publishing system
- Search
- Admin panel
- Engagement features

Phase 2:

- Analytics improvements
- UX improvements

- Search optimization

Phase 3:

- AI features
- Recommendation system

Phase 4:

- Multi-university expansion
 - Global scaling
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12. MVP Definition

Includes:

- Authentication
 - Profile management
 - Paper upload (multi-author)
 - Admin approval
 - Search
 - Reading & download
 - Reactions, comments, sharing
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13. Acceptance Criteria

Paper Upload:

- PDF $\leq$ 50MB
- Metadata required
- Supports multiple authors with order
- Status = pending

Search:

- Results within 2 seconds
- Filters operational

Authentication:

- Email verification required

- Invalid login blocked

Engagement:

- One reaction per user per paper
 - Comment and reply functional
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14. Product Philosophy

- Open access
- Simplicity
- Clean design
- Accessibility
- Scalability